



Database Retrieval

(NCBI, UniProt, Ensembl, PDB)

input.txt → output.fasta (single or multiple)



Sequence Alignment

(Pair-wise alignment & BLAST)

input.fasta (one sequence) → output.txt

BLASTN

BLASTP

BLASTX

TBLASTX

TBLASTN



Multiple Sequence Alignment (MSA)

(Clustal Omega)

input.fasta (multiple sequences) → output.fasta



Phylogenetic Tree Construction

(FastTree)

input.fasta (MSA output file) → output.newick



Visualize the Phylogenetic Tree

(ETE Toolkit)

input.newick (Phylogenetic tree) → output.png